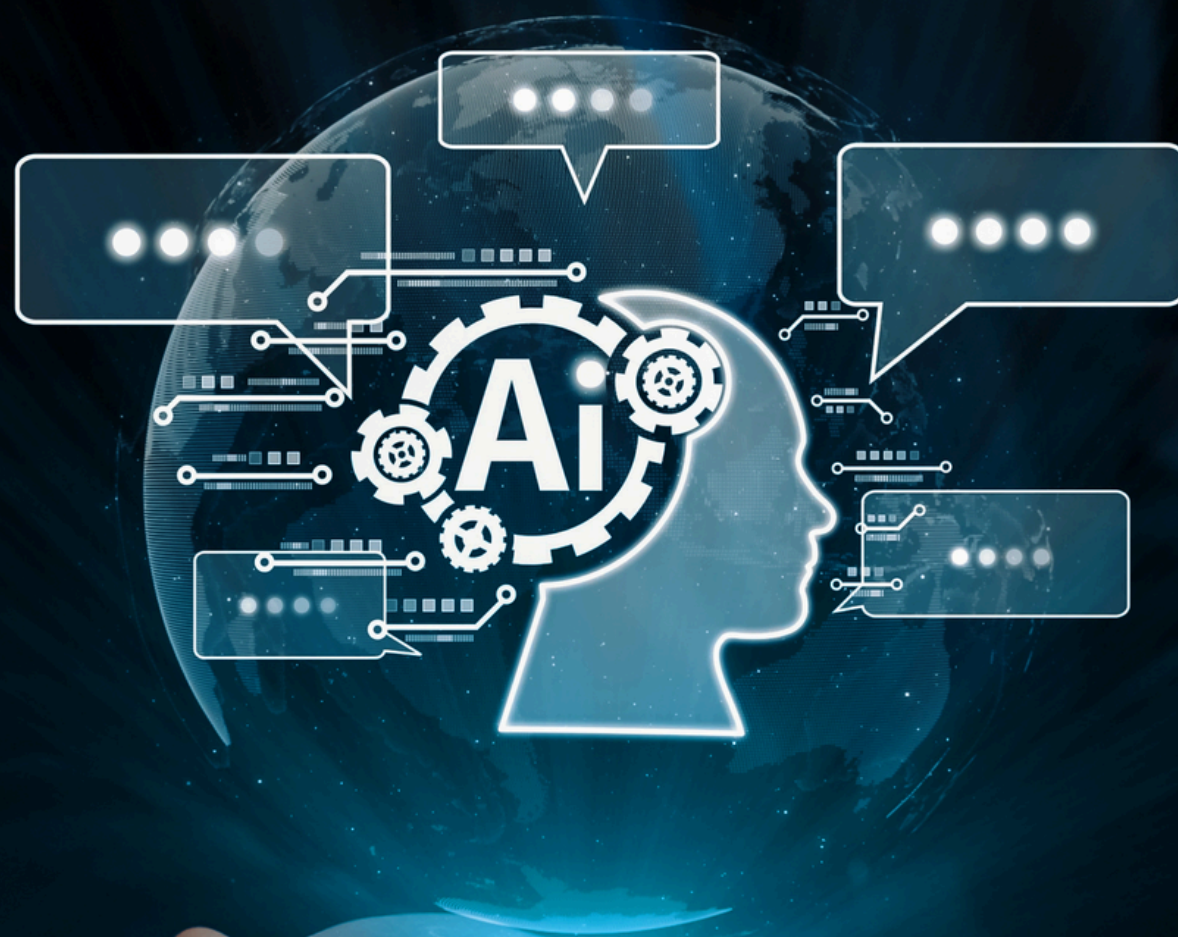




ADVANCED PROGRAM IN AI/ML

DURATION: 3 MONTHS



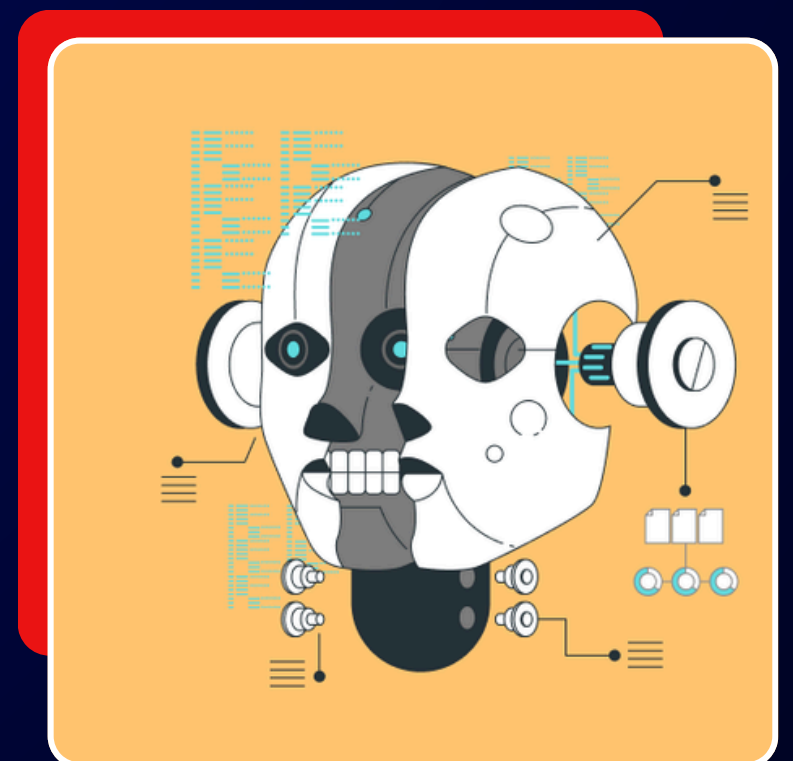
Class 1 – Fundamentals of AI / Background & Basics: Machine Learning Intuition

- Building Intuition Behind Computer Vision
- Human Eye and Computer Vision
- Camera Sensor and Photo Receptors
- Channels in an Image
- The Bored Cat Experiment
- Neural Computing and Real-time Models in Computer Vision
- Core Concept: Convolutions
- Receptive Field



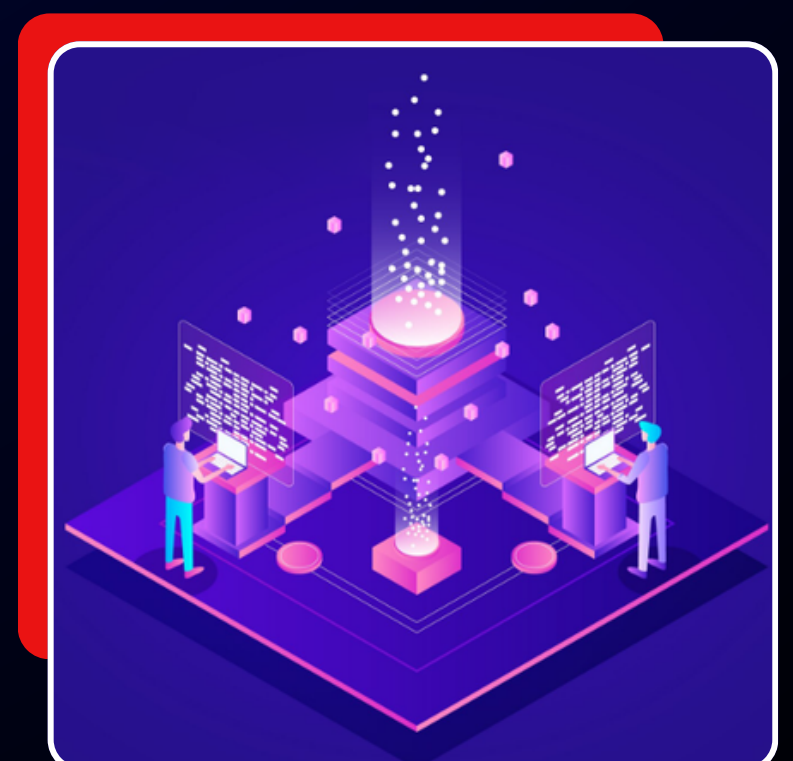
Class 2 – Prerequisite class: PyTorch 101:

- Introduction to PyTorch
- PyTorch vs TensorFlow
- Tensors
- Tensors and Numpy
- Sharing Memory for Performance
- Tensor Indexing
- Squeezing and Unsqueezing and other Operations
- Writing a NN from scratch in PyTorch:
 - Working with dataset
 - Building a network
 - Accessing the weights
 - Forward Function
 - Training
 - Loss
 - Gradients
 - Weight Updates
 - Batch Training
 - Epochs



Class 3 – Neural Networks and Architecture

- Understanding Neural Network Architecture
- Concepts from Class 1
- Importance of Adding Layers
- Receptive Field Calculation
- Convolutional Mathematics
- MaxPooling and its Importance
- Layer Count and Parameter Calculation
- Kernels in Layers
- Why 3x3 Kernel?
- Multi-Channel Convolution
- Our Network Now
- Assignment: Making a CV model on MNIST



Class 4 – Writing and Designing First Neural Network

- Creating Your First Neural Network
- Kernels and How They Work
- Convolution Models and Truth Behind 4 Blocks
- Convolutions and Strides
- Multichannel Convolution
- x1 Convolution and its Importance
- Modern Architectures
- Activation Functions
- Assignment: Bug Fixing on CNN Models



Class 5 – Architectural Basics

- Understanding Network Architectures
- Fully Connected Layers and Feed Forward Neural Networks
- VGG Net (Pre-2014)
- Modern Architectures and Designs (Post-2014)
- SoftMax Activation and its Role
- Architectural Ingredients
- MaxPooling In-Depth
- Batch Normalization Overview
- DropOut Overview
- Learning Rate Overview
- Batch Size Overview
- Assignment: Writing a Model with Few Parameters for Maximum Accuracy on MNIST



Class 6 – Coding a Real-Time Project: The Thought Processes

- Coding a Real-Time Computer Vision Project
- Code 1: Setting up Your Environment
- Code 2: Basic Skeleton of a Network
- Code 3: Lighter Model (First Model)
- Code 4: Adding and Improving Batch Normalization
- Code 5: Understanding and Implementing Regularization
- Code 6: Utilizing Global Average Pooling
- Code 7: Increasing Model Capacity and Parameter Analysis
- Code 8: Optimizing MaxPooling Placement
- Code 9: Image Augmentation Techniques
- Code 10: Experimenting with Learning Rates
- Receptive Field Calculations
- Assignment: State-of-the-Art Accuracy Model Training



Class 7 – Batch Normalization & Regularization

- Understanding Batch Normalization and Regularization
- Normalizing Input Data: Why and How
- Data Redistribution: Importance and Methods
- Weights with and without Normalization in a Network
- Understanding Batch Normalization and its Mathematics • Ghost Batch-Normalization
- Regularization: L1 & L2
- Implementing L1 & L2 Regularization in PyTorch
- Assignment: Adding Regularization to Models for Improved Performance



Class 8 – Advanced Convolutions and Concepts

- Exploring Advanced Convolutional Concepts
- Normal Convolutions
- Pointwise Convolutions
- Concept of Channels
- Addressing the Checkerboard Issue
- Atrous or Dilated Convolutions
- Dense Problems and Solutions
- Deconvolutions or Transpose Convolutions
- Understanding the Pixel Shuffle Algorithm
- Depthwise Separable Convolution
- Spatially Separable Convolution
- Grouped Convolutions
- Assignment: Working with Cifar 10 Dataset



Class 9 – Receptive Fields & Network Architectures

- Understanding Receptive Fields and Network Architectures
- Delving Deeper into Receptive Fields
- Receptive Fields Larger than Image Size
- VGG Architecture Analysis
- Receptive Field Changes with Added Layers
- Exploring Inception Architecture
- Inception V2 & V3 Architectures
- ResNet V1 & V2 Architectures
- ResNet V3 Architecture
- Assignment: Training ResNet on Cifar 100



Class 10 – Data Augmentation

- Enhancing Data and Validation Accuracy
- Relationship between Data and Validation Accuracy
- Data Augmentation Techniques
- PMDA and MMDA Augmentations
- Elastic Distortion and Cutout
- MixUp, RICAP, RMDA
- Patch Gaussian and Strategy
- Global Average Pooling
- GradCam for Diagnosing DNN
- Assignment: Applying Data Augmentation Strategies



Class 11 – Advanced Concepts in Training a Model & Learning Rates

- Optimizing Model Training and Learning Rates
- Understanding Learning Rate • Weight Update Strategies
- Constant Learning Rate Algorithms
- Adaptive Learning Rate Algorithms
- Stochastic Gradient Descent
- Gradient Perturbation
- Momentum & Nesterov Momentum
- RMSProp
- Adam Optimizer
- Implementing LRFinder for Finding Optimal Learning Rate
- Assignment: Implementing Real-time Learning Rate Finder



Class 12 – Super Convergence

- Achieving Very Fast Training of Neural Networks
- Background and Concepts
- Selecting an Optimal Starting Learning Rate
- Piecewise Constant Approach
- Hyper-parameter Grid Search
- Reduce Learning Rate on Plateau
- Warm-Up Strategies
- Cyclic Learning Rates for DNNs
- One Cycle Training
- Advantages of One Cycle Approach



Class 13 – Object Localization

- Understanding Object Localization
- Introduction and Comparison: Recognition vs. Detection
- Detection Approaches
- Recap of Techniques
- Obsolete Bounding Box Finding Methods
- Region Proposal Techniques
- Anchor Boxes: Concept and Computation
- Intersection over Union and Loss Functions
- Annotating a Dataset for Localization
- Assignment: Part 1 – Training YOLO



Class 14 – YOLO V2/3/4

- Exploring YOLO Versions
- YOLOV2 and Darknet-19
- Correct Loss Function for YOLOV2
- Understanding YOLOV3 and YOLOV4
- Reasons for Not Covering YOLOV5



Class 15 – RCNN Family: RCNN, Fast-RCNN, FasterRCNN & MaskRCNN

- Dive into the RCNN Family
- Introduction to RCNN Family
- RCNN and its Challenges
- Fast RCNN: Improvements Over RCNN
- Faster RCNN: Integrating Region Proposal Networks
- Region Proposal Networks in Detail
- Anchor Classification and Targeting
- Calculating Region Proposal Loss
- Mask R-CNN: Adding Mask Prediction Capability
- Recap of the RCNN Family



NLP To the Core

Class 1– Background and Basics of Modern NLP:

- Why do neural networks work so well?
- Kind of data we "actually" deal with in Real life.
- Feedforward Networks.
- Neurons and Rules.
- Mathematics. (Building Neural net in Excel)
- Architectural Ingredients (What/Why/Where we need?)



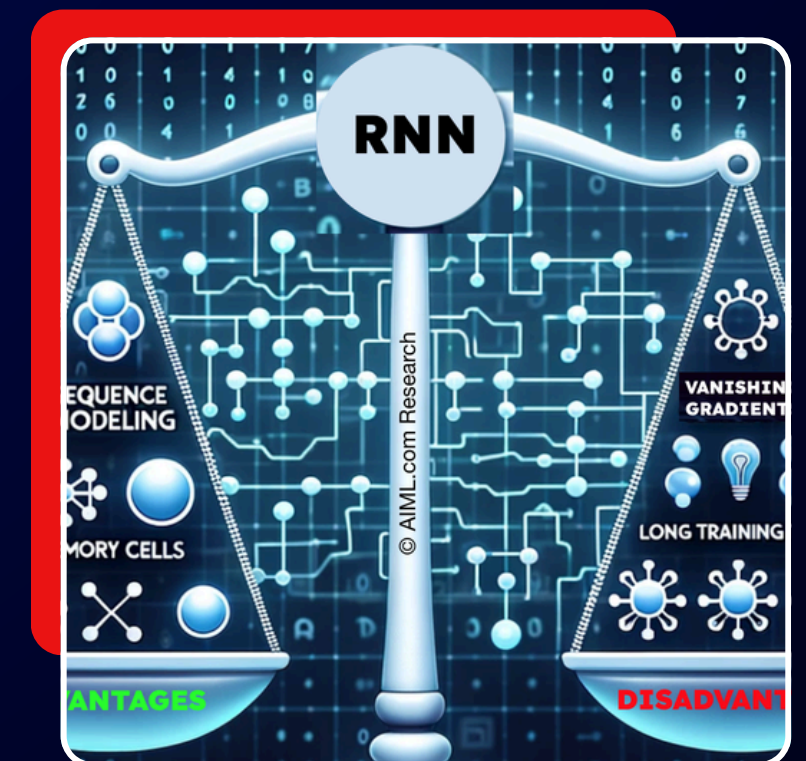
Session 2 – Back Propagation, Embeddings and Language Models

- Backpropagation
- Embeddings – An analogy from Visual Domain
- Embeddings
- Neural Networks
- Heavy Math – Forward Propagation
- Backpropagation
- Word Embeddings
- Modern Embeddings
- Word Embeddings vs Language Model
- Word2Vec



Session 3 – RNN are dead, and their renewed relevance

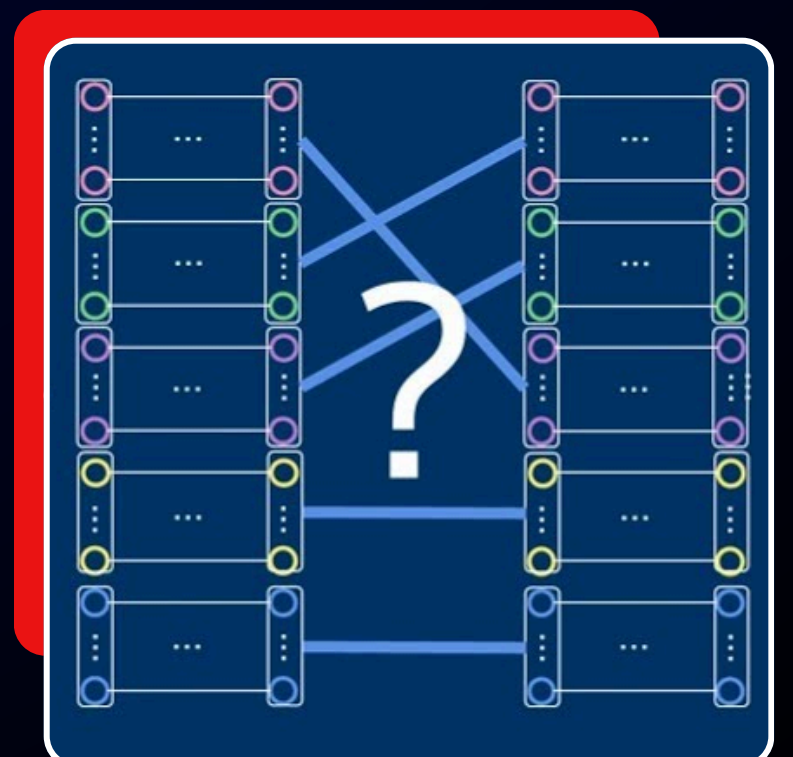
- Sequential Data
- Fully Connected Layers
- Recurrent Neural Networks
- RNN Cells
- Vanishing Gradients issue
- Training RNNs & LSTMs
- Transitioning from RNNs to LSTMs
- The tanh – Squashing Function
- Adding Memory to our recursion cells
- LSTM Cell & LSTM Core Concepts
- Sigmoid Gate, Forget Gate, Input Gate, Cell State ,Output Gate



Session 4– First Hands-on

Session 5 – GRUs, Seq2Seq and Introduction to Attention Mechanism

- RNN/LSTM Recap TanH – Squashing Function
- Sigmoid – Selection Gate
- LSTMs, LSTM Internals
- LSTM Variants, GRU
- Encoder=Decoder Architecture in RNN/LSTMs
- RNN/LSTM with an Attention Mechanism
- The Context Vector
- Computing Attention Weights and Context Vectors



AI Mastery Program

Month 1: Vision & Perception – “Seeing Like a Machine”

Computer Vision Track (Weeks 1–4)

Class 1 – Seeing Like a Machine

- How the Human Eye Inspires Computer Vision
- Sensors, Channels & the Science of Images
- The Curious Cat Experiment
- From Pixels to Perception: The Core Idea of Convolution

Class 2 – PyTorch Power-Up

- Essential PyTorch Concepts (Tensors, Ops & Gradients)
- Writing Your First Neural Network from Scratch

Class 3 – The Art of Neural Design

- Layers, Kernels & Receptive Fields
- MaxPooling Magic & Parameter Counting
- Building a Mini CV Model (MNIST Hands-On)

Class 4 – Your First Vision Model

- Decoding Convolution Blocks & Strides
- 1x1 Convolutions & Why They Matter
- Modern Architectures in Action



Month 2: Deep Networks & Smart Training

Computer Vision (Weeks 5–6)

Class 5 – Blueprint of Deep Networks

- From VGG to Modern CNNs
- BatchNorm, Dropout & Learning Rate Secrets
- Hands-On: Compact Model for High Accuracy

Class 7 – Train Smarter, Not Harder

- Regularization, Fine-Tuning & Transfer Learning
- Experimenting with Learning Rates & Schedulers

NLP Track (Weeks 7–8)

Class 1 – The Language of Machines

- Why Neural Networks Understand Words
- Real-World Data: From Text to Meaning
- Building Neural Intuition (Neurons, Rules & Flow)
- Hands-On: Mini Neural Net in Excel

Class 3 – The Rise (and Return) of RNNs

- Sequential Thinking & RNN Fundamentals
- Vanishing Gradients & How LSTMs Fixed It
- Inside the LSTM Cell: Memory, Gates & Flow

Class 6 – From Code to Creation

- Building a Real-Time Vision Project
- Data Augmentation & Model Optimization Tricks

Class 8 – Beyond Basic Convolutions

- Depthwise, Dilated & Grouped Convolutions
- Understanding Pixel Shuffle & Deconvolution

Class 2 – How Machines Learn Language

- Backpropagation Simplified
- What Are Word Embeddings?
- From Word2Vec to Modern Embeddings

Month 3: Attention & Agent Foundations

NLP Track (Weeks 9–10)

Class 4 – Hands-On with Sequence Models

- Implementing & Visualizing RNN/LSTM Behavior
- Real-World Application Exercise

Class 5 – Attention Changes Everything

- GRU Simplified: Smarter, Lighter RNNs
- Encoder–Decoder Architecture
- Birth of Attention Mechanism
- How Context Vectors Transform Understanding



AI Agent Mastery (Weeks 11–12)

Session 1 – Transformer Basics

- Tokens, Embeddings & Attention
- Self-Attention Mechanics & Positional Encoding

Session 2 – LLM Internals

- Architecture of Modern LLMs
- Tokenization & Vocabulary
- Pretraining vs Fine-Tuning
- Output Decoding

Session 3 – Introduction to Agentic AI

- What Are Agents?
- Autonomy, Decision-Making & Real-World Examples
- Role of LLMs in Agents

Session 4 – Model Context Protocol (MCP)

- Context Windows, Memory Handling & State Management
- Communication Between Agents
- Context-Aware Reasoning



Month 4: Advanced Agents & Deployment

AI Agent Mastery (Weeks 13–16)

Session 5 – Reasoning & Planning

- Chain-of-Thought Frameworks
- Multi-Step Decision Making
- Task Decomposition

Session 6 – Agentic Architecture

- Core Components & Workflow Design
- Input, Memory & Action Loops
- Modular Agent Systems

Session 7 – Memory Systems

- Short-Term vs Long-Term Memory
- Episodic & Semantic Memory
- Memory Retrieval Mechanisms
- Vector Databases

Session 8 – Tool Use & RAG 2.0

- Using External APIs
- Document & Knowledge Retrieval
- Retrieval-Augmented Generation
- Context Injection

Session 9 – Hybrid Planning

- Combining Symbolic & Neural Approaches
- Adaptive Task Handling
- Real-Time Execution

Session 10 – Multi-Agent Systems

- Collaboration & Communication
- Task Distribution & Conflict Resolution
- Swarm Intelligence

Session 11 – Browser Agents

- Web Automation & Information Extraction
- Browser API Integration

Session 12 – Computer Agents

- OS Interaction, File & App Automation
- Command Execution & Environment Awareness

Session 13 – Task Orchestration & Deployment (Optional Platform)

- Dynamic Scheduling & Workflow Coordination
- Resilient Execution & Real-Time Adaptation
- Cloud Setup, Scaling, Monitoring & Optimization



Bangalore

Bhubaneswar

Dubai

DV Data & Analytics Private Limited

Bangalore Center

**DV Data & Analytics Bangalore
Private Limited**

#52, 2nd Floor:
Malleshpalya Maruthinagar
Bengaluru.
Bangalore 560075, India

Phone: (+91) 9019 030 033
(+91) 8095 881 188
Email: info@dvanalyticsmds.com

Bhubaneswar Center

**DV Data & Analytics Private Limited
Bhubaneswar**

Plot No A/7 :
Adjacent to Maharaja Cine Complex,
Bhoinagar, Acharya Vihar
Bhubaneswar 751022, India

Phone: (+91) 8095 881 188
(+91) 8249 430 414
Email: info@dvanalyticsmds.com

Dubai Center

Agentify AI Global

Villa 335
Murooj al Furzan East 1
Near Gate 3 Jabel Ali First
Al Furjan, Dubai, UAE

Phone: +971 52 553 3211
Email: Sales@agentifyaiglobal.com